

Day 99 Project

European Option Pricing under Jump-Diffusion

1 Introduction

This project extends option pricing beyond the Black-Scholes framework by incorporating jumps in asset prices.

We use the Merton Jump-Diffusion model and Monte Carlo simulation to price European options.

2 Model Dynamics

The stock price follows:

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t(J - 1)dN_t$$

where:

- W_t is Brownian Motion
- N_t is a Poisson process with intensity λ
- J represents jump size

3 Simulation Approach

Monte Carlo pricing involves:

1. Simulating price paths under jump-diffusion

2. Computing option payoff at maturity
3. Averaging and discounting the payoff

4 Jump Component

$$N_t \sim \text{Poisson}(\lambda \Delta t)$$

$$J = \exp(\mu_J + \sigma_J Z)$$

5 Option Payoff

For a European call option:

$$\max(S_T - K, 0)$$

6 Key Insights

- Jump risk significantly impacts option pricing
- Standard models underestimate tail risk
- Monte Carlo enables flexible modeling

7 Conclusion

This project demonstrates how to incorporate real-world market behavior into option pricing using stochastic models and simulation techniques.